

Certificate Chain / Trust Proof

Document: PEPPOL AP Certificate Chain & Trust Validation **Service Provider:** AL MERAK TAX CONSULTANT L.L.C (E-Numerak) **Date:** 2026-06-20

1. Purpose

Demonstrate that our Access Point certificate chains up to a trusted PEPPOL root, and that our AP **validates** the trust chain (and revocation status) of every remote certificate before transmitting or accepting a message. This is the "trust proof" required for ASP registration.

2. The PEPPOL G3 trust chain

A PEPPOL AP certificate is a 3-level chain:

PEPPOL Root CA - G3

(root, self-signed)

└─ PEPPOL ACCESS POINT CA - G3

(intermediate, issuing CA)

└─ CN=PAE001147 (our AP cert)

(leaf / end-entity)

- **Test** chain issuer: PEPPOL ACCESS POINT TEST CA - G3 ("FOR TEST ONLY").
- **Production** chain issuer: PEPPOL ACCESS POINT CA - G3 .

Our current leaf (test):

Field	Value
Subject	CN=PAE001147, OU=PEPPOL TEST AP, O=AL MERAK TAX CONSULTANT L.L.C, C=AE
Issuer	CN=PEPPOL ACCESS POINT TEST CA - G3, OU=FOR TEST ONLY, O=OpenPEPPOL AISBL, C=BE
Not After	2028-05-20

3. Assembling the chain (after issuance)

Once OpenPEPPOL returns the issued leaf certificate:

```
# 1. Convert the issued cert to PEM (if returned as DER/.cer)
openssl x509 -inform DER -in peppol-ap-prod.cer -out leaf.pem # skip if already PEM

# 2. Build the full-chain file: leaf -> intermediate -> root
cat leaf.pem peppol-ap-ca-g3.pem peppol-root-ca-g3.pem > peppol-ap-prod-fullchain.pem

# 3. Verify the chain end-to-end
openssl verify -CAfile peppol-root-ca-g3.pem \
  -untrusted peppol-ap-ca-g3.pem leaf.pem
# expected: leaf.pem: OK

# 4. Package key + chain into a PKCS#12 for the runtime keystore
openssl pkcs12 -export \
  -inkey certs/production/peppol-ap-prod.key \
  -in leaf.pem \
  -certfile peppol-ap-ca-g3.pem \
  -out certs/production/ap-prod.p12.pfx \
  -name "peppol-ap-prod"
```

The PEPPOL intermediate + root certificates are published by OpenPEPPOL and are already bundled in our trust store (see §4).

4. How our AP proves trust (code)

Trust/revocation validation is implemented in [services/as4/cert_validation.py](#), with the bundled PEPPOL anchors in [services/as4/trust_anchors/](#).

Outbound (before we send): the recipient AP certificate returned by SMP is validated by `validate_recipient_cert()` — chain build to a bundled PEPPOL anchor **and** CRL revocation check. If it fails, the message is **not transmitted**. This was certified in the eDelivery testbed (TC2A.4, invalid-cert handling) and the PINT-AE Business Document Validation suites — see the batch summary lines `REJECTED-INVALID-CERT`.

Inbound (before we accept): the signer certificate on a received AS4 message is run through the same chain/revocation gate (controlled by `PEPPOL_AS4_VERIFY_SIGNER_TRUST`). Untrusted signers are rejected — certified in the PKI G3 suite (invalid-cert reception).

Evidence already on file: - [TestBedReport-PAE001147-20260609T122213.pdf](#) — eDelivery Standalone AP 6/6 (incl. TC2A.4). - [TestBedReport-PAE001147-20260609T135555.pdf](#) — PKI G3 suite 3/3 (valid + invalid cert).

5. Checklist

- ☐ Receive issued production leaf certificate from OpenPEPPOL.
- ☐ Download PEPPOL production intermediate + root (G3).
- ☐ Build & `openssl verify` the full chain (must say `OK`).
- ☐ Confirm leaf public key matches `peppol-ap-prod.key` (modulus check).
- ☐ Export password-protected PKCS#12 for the keystore.
- ☐ Add production anchors to `services/as4/trust_anchors/` if not present.

```
# Modulus match check (leaf cert vs our private key) – both hashes must be equal
openssl x509 -noout -modulus -in leaf.pem | openssl md5
openssl rsa -noout -modulus -in certs/production/peppol-ap-prod.key | openssl md5
```